THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · HARD LEVEL UNIT TEST SET 1

Chapters 1–5: **Patterns in Mathematics, Lines and Angles, Number Play, Data Handling and Presentation, Prime Time** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. At a party, 15 people each hug every other person exactly once. Number of hugs: 1

(a) 105 **(b)** 112 **(c)** 120 **(d)** 210

2. A clock shows 2 o'clock. After 3 hours, the smaller angle between the clock hands is: 1

(a) 150 **(b)** 210 **(c)** 90 **(d)** 60

3. How many steps does the Kaprekar routine take to reach 6174 from 3141? 1

(a) 4 **(b)** 6 **(c)** 7 **(d)** 5

4. The total of the table, after removing the largest value, is: 1

Item	Number
Banana	75
Mango	25
Orange	15
Apple	55
Grapes	45

(a) 215 **(b)** 140 **(c)** 145 **(d)** 200

5. The smallest 4-digit number divisible by 13 is: 1

(a) 1001 **(b)** 1002 **(c)** 1014 **(d)** 988

6. If $1 + 2 + 3 + \cdots + n = 703$, then n is: 1

(a) 19 **(b)** 38 **(c)** 36 **(d)** 37

7. A wheel turns 3 full turns and then a further quarter turn. The total angle in degrees is: 1

(a) 1080 **(b)** 1170 **(c)** 1260 **(d)** 1350

8. The hailstone (Collatz) sequence of 25 has how many terms (including 25 and 1)? 1

(a) 23 **(b)** 22 **(c)** 24 **(d)** 25

9. One symbol of a pictograph stands for 10. Class A has 12 symbols and its total is 3 times Class B. How many symbols does Class B have? 1

(a) 9 **(b)** 5 **(c)** 4 **(d)** 36

10. The number of composite numbers from 1 to 30 (both included) is: 1

(a) 19 **(b)** 20 **(c)** 18 **(d)** 21

11. The sum of the first 30 odd numbers is 900. The sum of the odd numbers from 21 to 59 is: 1

(a) 779 **(b)** 800 **(c)** 400 **(d)** 890

12. A spinner starts pointing North and turns clockwise 270° then anticlockwise 90°. The final angle clockwise from North is: 1

(a) 270 **(b)** 190 **(c)** 170 **(d)** 180

13. **Assertion (A):** In the sequence $3, 7, 11, 15, \ldots$ the next term is 19. 1

Reason (R): Each term of this sequence is obtained by adding 4 to the term before it.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

14. **Assertion (A):** An angle of 135° is an obtuse angle. 1

Reason (R): An obtuse angle is more than 90° and less than 180°.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

15. **Assertion (A):** The largest number that can be made from the digits 3, 0, 7, 5 (each used once) is 7530. 1

Reason (R): To make the largest number, arrange the digits in descending order.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. 14 cricket teams play a league in which every team plays every other team once. Find the number of matches. 2

17. Find the smaller angle between the clock hands at 4 o'clock, and the angle the hour hand turns through from 4 o'clock to 6 o'clock. 2

18. How many steps does the Kaprekar routine need for 1234? Write the chain. 2

19. In a survey of 150 people, 62 like tea, 48 like coffee, and the rest like milk. Find the number liking milk and show data in a table. 2

20. Find the smallest number divisible by 8, 12 and 18. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. In a pattern the 1st term is 14, the 2nd is 23, and so on by adding 9. Which term of the pattern equals 365? Find also the term just before it. 3

22. At 1 o'clock the hands of a clock make a smaller and a larger angle. Find both and say which one is a reflex angle (if any). 3

23. Find a 4-digit palindrome of the form $abba$ with $a = 1$ and $b = 5$. Show that it is divisible by 11. 3

24. The table shows two weeks of sales (units). Find the week with the larger total and by how much. 3

Day	Week 1	Week 2
Mon	13	15
Tue	20	19
Wed	11	14
Thu	17	12

OR

Three traffic lights change every 48, 72 and 108 seconds. If they change together at 8:00 am, after how many seconds do they next change together and at what time? 3

25. Sums of consecutive odd numbers give squares. Find the sum of the odd numbers from 1 to 39. Then find the sum of the odd numbers from 41 to 49. 3

OR

At 2:00 the minute hand starts turning clockwise and turns through $90°$. How many minutes pass, how many right angles does it turn, and what time does the clock then show? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** Ravi makes a row of squares with matchsticks. One square needs 4 sticks, two squares joined side by side need 7 sticks, and three squares need 10 sticks. He keeps adding one square at a time. 5

(i) How many sticks are needed for 4 squares?
(ii) How many sticks does each new square add?
(iii) How many sticks are needed for 10 squares?
(iv) Ravi has exactly 40 sticks. How many squares can he make in a row? Show how you found it.

27. A palindromic number of 5 digits is formed as $abcba$ where $a = 1$, $b = 2$, $c = 3$. Write the number, find its digit sum, check if it is divisible by 9, and find the smallest number to add to make it divisible by 9. 5

OR

A bar graph has scale 1 cm = 12 units. Bars A, B, C have heights 3 cm, 6 cm, 9 cm. The value of D equals A + B - C. Find the values of A, B, C and D, and the height of bar D. 5

END OF QUESTION PAPER — Total: 50 Marks

MASTERMIND